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Subject: Programming with Java

ASSIGNMENT No : 24

TITLE : Develop a Java GUI application integrated with JDBC to perform database operations.

Problem Statement

Develop a Java GUI application integrated with JDBC to perform database operations.

Learning Objectives

To integrate Java GUI applications with databases using JDBC.

Theory

Java Swing provides graphical user interface components such as buttons, labels, text fields, tables, and forms, while JDBC enables database connectivity. Combining Swing with JDBC allows developers to create interactive desktop applications capable of performing database operations through a graphical interface. Users can enter information using GUI components, which is then processed and stored in the database using SQL queries. Retrieved records can be displayed in tables or text fields for easy viewing and modification. This integration improves user experience by eliminating command-line interaction and forms the foundation of desktop applications such as inventory management systems, library management systems, and student management systems.

Output Screenshots

General Login

Staff Login

System Login

Username

Enter Username

Password

Enter Password

Login

General Login

Staff Login

System Login

Username

admin

Password

●●●●●●●●

Login

Login Successful! Welcome, admin.

Library Management

book1

Add Book

Message

i

Book Added

OK

INDEX CODE

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Hospital Management System - Portals</title>
7   <style>
8     * {
9       box-sizing: border-box;
10      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
11      margin: 0;
12      padding: 0;
13    }
14
15    body {
16      background-color: #eef2f5;
17      display: flex;
18      justify-content: center;
19      align-items: center;
20      height: 100vh;
21    }
22
23    .container {
24      background-color: #ffffff;
25      width: 400px;
26      padding: 30px;
27      border-radius: 8px;
28      box-shadow: 0 4px 15px rgba(0, 0, 0, 0.1);
29    }
30
31    .tabs {
32      display: flex;
33      margin-bottom: 25px;
34      border-bottom: 2px solid #e0e0e0;
35    }
36
37    .tab-btn {
38      flex: 1;
39      padding: 10px;
40      text-align: center;
41      background: none;
42      border: none;
43      font-size: 15px;
```

```
44         font-weight: bold;
45         color: #777;
46         cursor: pointer;
47         outline: none;
48     }
49
50     .tab-btn.active {
51         color: #007bff;
52         border-bottom: 3px solid #007bff;
53     }
54
55     .form-group {
56         margin-bottom: 20px;
57     }
58
59     .form-group label {
60         display: block;
61         font-size: 14px;
62         color: #333;
63         margin-bottom: 6px;
64     }
65
66     .form-group input {
67         width: 100%;
68         padding: 10px;
69         border: 1px solid #ccc;
70         border-radius: 4px;
71         font-size: 14px;
72     }
73
74     .btn {
75         width: 100%;
76         padding: 12px;
77         background-color: #007bff;
78         color: white;
79         border: none;
80         border-radius: 4px;
81         font-size: 16px;
82         font-weight: bold;
83         cursor: pointer;
84         transition: background 0.3s;
85     }
86
87     .btn:hover {
88         background-color: #0056b3;
```

```

89     }
90
91     .form-section {
92         display: none;
93     }
94
95     .form-section.active {
96         display: block;
97     }
98
99     .status-msg {
100         margin-top: 15px;
101         font-size: 14px;
102         text-align: center;
103         font-weight: bold;
104     }
105 </style>
106 </head>
107 <body>
108
109 <div class="container">
110     <div class="tabs">
111         <button class="tab-btn active" onclick="switchTab('user-
login', this)">General Login</button>
112         <button class="tab-btn" onclick="switchTab('staff-
login', this)">Staff Login</button>
113     </div>
114
115     <!-- General User Login Form -->
116     <div id="user-login" class="form-section active">
117         <h2 style="text-align: center; margin-
bottom: 20px; color: #333;">System Login</h2>
118         <div class="form-group">
119             <label for="username">Username</label>
120             <input type="text" id="username" placeholder="Enter Username">
121         </div>
122         <div class="form-group">
123             <label for="user-password">Password</label>
124             <input type="password" id="user-
password" placeholder="Enter Password">
125         </div>
126         <button class="btn" onclick="handleUserLogin()">Login</button>
127         <div id="user-msg" class="status-msg"></div>
128     </div>
129

```

```

130     <!-- Hospital Staff Login Form -->
131     <div id="staff-login" class="form-section">
132         <h2 style="text-align: center; margin-
bottom: 20px; color: #333;">Hospital Staff Portal</h2>
133         <div class="form-group">
134             <label for="staff-id">Staff ID</label>
135             <input type="text" id="staff-
id" placeholder="e.g., DOC101 or NUR201">
136         </div>
137         <div class="form-group">
138             <label for="staff-password">Password</label>
139             <input type="password" id="staff-
password" placeholder="Enter Password">
140         </div>
141         <button class="btn" onclick="handleStaffLogin()">Login to Portal</
button>
142         <div id="staff-msg" class="status-msg"></div>
143     </div>
144 </div>
145
146 <script>
147     function switchTab(tabId, element) {
148         document.querySelectorAll('.form-
section').forEach(el => el.classList.remove('active'));
149         document.querySelectorAll('.tab-
btn').forEach(el => el.classList.remove('active'));
150         document.getElementById(tabId).classList.add('active');
151         element.classList.add('active');
152     }
153
154     function handleUserLogin() {
155         const user = document.getElementById('username').value;
156         const pass = document.getElementById('user-password').value;
157         const msg = document.getElementById('user-msg');
158
159         if (user === "admin" && pass === "admin123") {
160             msg.style.color = "green";
161             msg.innerText = "Login Successful! Welcome, " + user + ".";
162         } else {
163             msg.style.color = "red";
164             msg.innerText = "Invalid Username or Password. Access Denied.";
165         }
166     }
167
168     function handleStaffLogin() {

```

```
169     const id = document.getElementById('staff-id').value;
170     const pass = document.getElementById('staff-password').value;
171     const msg = document.getElementById('staff-msg');
172
173     if (id === "DOC101" && pass === "docpass") {
174         msg.style.color = "green";
175
176         msg.innerText = "Access Granted: Welcome, Dr. " + id + ". Redirecti
ng to Doctor Dashboard.";
177     } else if (id === "NUR201" && pass === "nurpass") {
178         msg.style.color = "green";
179
180         msg.innerText = "Access Granted: Welcome, Nurse " + id + ". Redirec
ting to Nursing Station.";
181     } else {
182         msg.style.color = "red";
183
184         msg.innerText = "Authentication Failed: Invalid Staff ID or Passwor
d.";
185     }
186 }
187 </script>
188 </body>
189 </html>
```

CODE 1

```
1 package Exp24;
2
3 import javax.swing.*;
4 import java.sql.*;
5
6 public class Main {
7     public static void main(String[] args) {
8         JFrame f = new JFrame("Library Management");
9
10        JTextField name = new JTextField();
11        JButton add = new JButton("Add Book");
12
13        name.setBounds(50, 50, 200, 30);
14        add.setBounds(100, 100, 120, 30);
15
16        f.add(name);
17        f.add(add);
18
19        add.addActionListener(e -> {
20            try {
21                Connection con = DriverManager.getConnection(
22                    "jdbc:mysql://localhost:3306/hospital_db",
23                    "root", "devansh@2007");
24
25                PreparedStatement ps = con.prepareStatement(
26                    "INSERT INTO books(name) VALUES(?)");
27
28                ps.setString(1, name.getText());
29                ps.executeUpdate();
30
31                JOptionPane.showMessageDialog(f, "Book Added");
32                con.close();
33            } catch (Exception ex) {
34                JOptionPane.showMessageDialog(f, ex.getMessage());
35            }
36        });
37
38        f.setSize(320, 200);
39        f.setLayout(null);
40        f.setVisible(true);
41    }
42 }
```


EXERCISE 1

```
1 package Exercise.Exp24;
2
3 import javax.swing.*;
4 import java.sql.*;
5
6 public class Exp1 {
7     public static void main(String[] args) {
8         JFrame f = new JFrame("Library");
9         JTextField book = new JTextField();
10        JButton add = new JButton("Add");
11
12        book.setBounds(50, 40, 200, 30);
13        add.setBounds(100, 90, 80, 30);
14
15        f.add(book);
16        f.add(add);
17
18        add.addActionListener(e -> {
19            try {
20                Connection con = DriverManager.getConnection(
21                    "jdbc:mysql://localhost:3306/hospital_db",
22                    "root", "devansh@2007");
23
24                PreparedStatement ps = con.prepareStatement(
25                    "INSERT INTO books(name) VALUES(?)");
26                ps.setString(1, book.getText());
27                ps.executeUpdate();
28
29                JOptionPane.showMessageDialog(f, "Book Added");
30                con.close();
31            } catch (Exception ex) {
32                JOptionPane.showMessageDialog(f, "Error");
33            }
34        });
35
36        f.setSize(300, 180);
37        f.setLayout(null);
38        f.setVisible(true);
39    }
40 }
```

EXERCISE 2

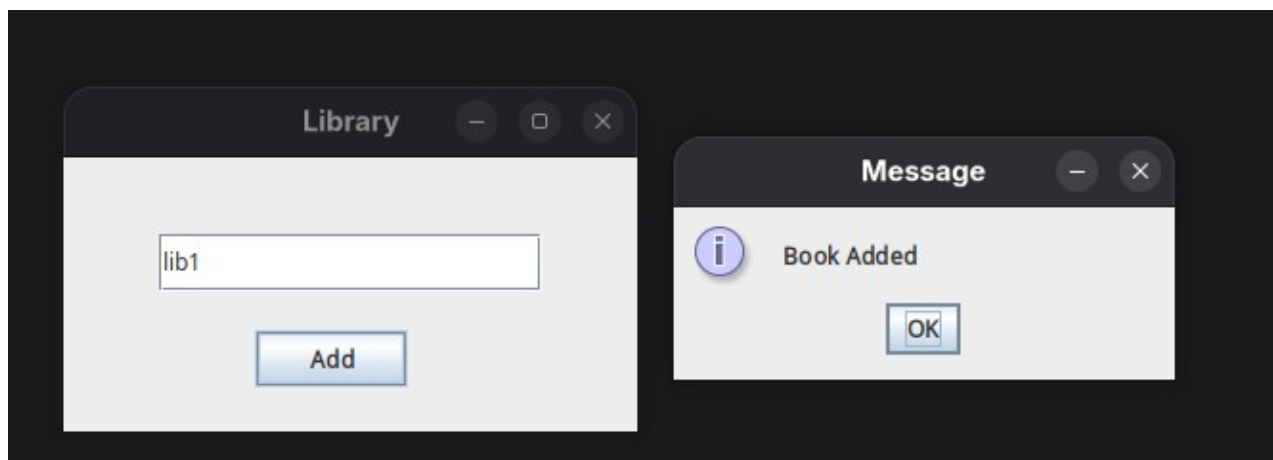
```
1 package Exercise.Exp24;
2
3 import javax.swing.*;
4 import java.sql.*;
5
6 public class Exp2 {
7     public static void main(String[] args) {
8         JFrame f = new JFrame("Library");
9         JTextField book = new JTextField();
10        JButton add = new JButton("Add");
11
12        book.setBounds(50, 40, 200, 30);
13        add.setBounds(100, 90, 80, 30);
14
15        f.add(book);
16        f.add(add);
17
18        add.addActionListener(e -> {
19            try {
20                Connection con = DriverManager.getConnection(
21                    "jdbc:mysql://localhost:3306/hospital_db",
22                    "root", "devansh@2007");
23
24                PreparedStatement ps = con.prepareStatement(
25                    "INSERT INTO books(name) VALUES(?)");
26                ps.setString(1, book.getText());
27                ps.executeUpdate();
28
29                JOptionPane.showMessageDialog(f, "Book Added");
30                con.close();
31            } catch (Exception ex) {
32                JOptionPane.showMessageDialog(f, "Error");
33            }
34        });
35
36        f.setSize(300, 180);
37        f.setLayout(null);
38        f.setVisible(true);
39    }
40 }
```

SQL CODE:

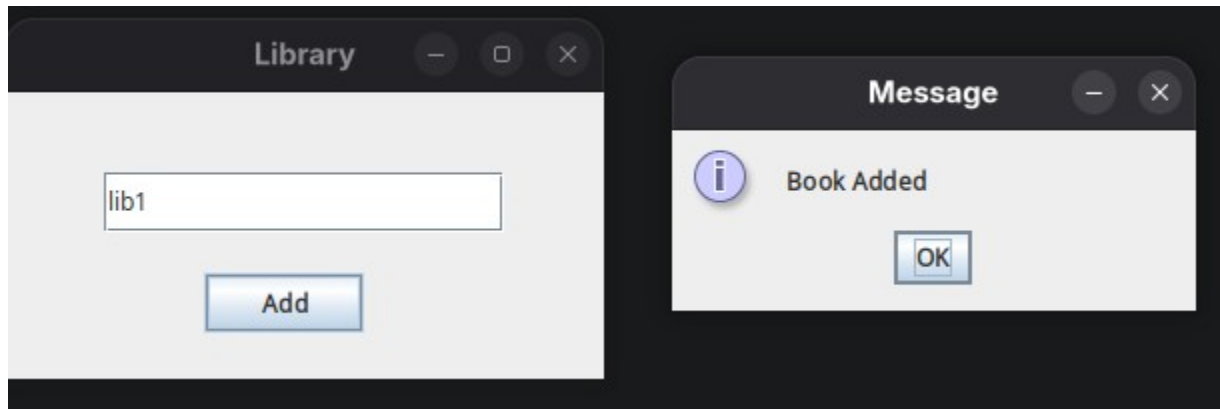
```
1 USE hospital_db;
2
3 CREATE TABLE books (
4     id INT AUTO_INCREMENT PRIMARY KEY,
5     name VARCHAR(100)
6 );
7
8 CREATE TABLE book_issue (
9     book_id INT,
10    student_name VARCHAR(100),
11    issue_date DATE,
12    return_date DATE
13 );
14
```

Exercise

Q1. Create a Library Management System GUI connected with database.



Q2. Create a Book Issue Tracking System GUI application using Java Swing connected to a database. The system should store book ID, student name, issue date, and return date, and allow record management.



Conclusion

Thus, the experiment successfully demonstrated the integration of Java Swing GUI with JDBC for performing database operations. The GUI provides an interactive way to enter and manage data, while JDBC enables communication with the database. This integration helps in developing user-friendly desktop applications with efficient database management.